Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD model (Prova 01 remediation, Q3)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Notation (as in Prova 01; Benigno 2015). Lower-case letters are logs (deviations from the steady state); bars denote the long run; $\Omega \equiv \sigma^{-1} + \eta$; $\rho \equiv -\ln\beta$, so $-\sigma\ln\beta = \sigma\rho$.

$$y = \bar{y} + (g - \bar{g}) - \sigma[i - (\bar{p} - p)] - \sigma\ln\beta, \tag{AD}$$

$$p - p^e = \kappa(y - y_n), \tag{AS}$$

$$y_n = \tfrac{1+\eta}{\Omega}\, a + \tfrac{\sigma^{-1}}{\Omega}\, g - \tfrac{1}{\Omega}\,\mu, \tag{YN}$$

$$\bar{y} = \tfrac{1+\eta}{\Omega}\,\bar{a} + \tfrac{\sigma^{-1}}{\Omega}\,\bar{g} - \tfrac{1}{\Omega}\,\bar{\mu}, \tag{YB}$$

$$y_e = \tfrac{1+\eta}{\Omega}\, a + \tfrac{\sigma^{-1}}{\Omega}\, g, \tag{YE}$$

$$L = \tfrac{1}{2}(y - y_e)^2 + \tfrac{\phi}{2\kappa}(p - p^e)^2, \qquad \phi \geq 0 \qquad \text{(loss of the central bank)}. \tag{LF}$$

a, μ, g are current productivity, (log) markup and public spending; $\bar{a}, \bar{\mu}, \bar{g}$ their long-run values. Parameters: $\sigma, \eta, \kappa > 0$. Unless stated, the economy starts at an efficient steady state ($y = y_n = y_e = \bar{y}$, $p = p^e = \bar{p}$, $g = \bar{g}$; a subsidy offsets the steady-state markup, so $\mu, \bar{\mu}$ are deviations from it), the bank sets i, and p^e and $\bar{p}$ do not move. "Shock" means a change $d(\cdot)$ from that point.

Block 1 — Exam level: Benigno's own experiments, Lista 7 and Carlin–Soskice

(Benigno §§6, 7, 11 • 40 points)

Question 1. (10 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §6.1–§6.2, Figs. 6–7, p. 512 — the article has no exercises; the statement below is ours. The article treats a productivity* gain*; here the sign is reversed.* Redo the two experiments of those sections for a productivity *fall*, $da < 0$.

Context from the source (added by this list, from Benigno 2015, pp. 508–509, 512–513 and 522): Benigno's numbering of the cover equations: (AS) is his (17), p. 508, repeated as (20), p. 509; (YN) is (15), p. 508; (YB) is the unnumbered $\bar{y}_n$ of §4.2, p. 508 (the cover writes $\bar{y}$); (YE) is (19), p. 509; (AD) is (21), p. 509, with his consumption-tax term $(\bar{\tau}_c - \tau_c)$ set to zero; (LF) is (35), p. 522. *Fig. 6* (temporary gain, $a \uparrow$, $\bar{a}$ fixed): AS shifts down/right to AS′, through (y_n', p^e); AD does not move, since it does not depend on current productivity; the equilibrium moves from E to E', with lower prices, higher output and a negative output gap; the efficient level of output rises in the same proportion as the natural level. Lowering i shifts AD up to AD″, reaching E'': stable prices and zero output gap. *Fig. 7* (permanent gain, $a \uparrow$ and $\bar{a} \uparrow$): AS shifts to AS′ as in Fig. 6, and AD shifts up to AD′ because the future natural level of consumption rises; the new equilibrium E' has $p = p^e$ and $y = y_n'$ without any policy intervention. p. 513: in the initial equilibrium the nominal rate equals the natural real rate of interest,

$$i = r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n) + \sigma^{-1}(g - \bar{g}) + (\bar{\tau}_c - \tau_c),$$

and a permanent productivity gain leaves r_n unchanged because $\bar{y}_n$ and y_n rise in the same proportion.

What the article states (p. 512): "First, we analyze the case in which the economy undergoes a temporary productivity gain, meaning that productivity rises in the short run but does not vary in the long run." (§6.1) and "Fig. 7 analyzes the case of a permanent productivity shock." (§6.2)

a) (4 points) The fall is temporary ($d\bar{a} = 0$) and i stays at its initial level. Which curve moves, in which direction and why? Find dy, dp, $d(y - y_n)$ and $d(y - y_e)$ in closed form, with signs.

Solution:

b) (3 points) Find the change in i that keeps $p = p^e$. Is it a cut or a rise? What does it cost in terms of $y - y_e$?

Solution:

c) (3 points) Now the fall is permanent ($d\bar{a} = da$). Answer (a) and (b) again and explain the difference.

Solution:

Question 2. (10 points) *Source: built on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §6.3, Fig. 8, p. 513 — the article has no exercises; the statement below is ours. Item (a) redoes the article's experiment; items (b)–(c), the markup analogue, are added by this list.*

Context from the source (added by this list, from Benigno 2015, p. 513): *Fig. 8* (an increase in long-run productivity, $\bar{a} \uparrow$, current a fixed): AS does not move, since current productivity is unchanged; AD shifts up/right to AD$'$, because households expect higher consumption in the future and, to smooth it, want to raise consumption today. The equilibrium moves from E to E': higher prices, and output above both its natural and its efficient level. Raising the nominal interest rate brings AD back to the initial point E. The article reads the same experiment as optimism (or, with $\bar{a} \downarrow$, pessimism) about future productivity.

What the article states (p. 513): "Consider now the case of an expected increase in long-run productivity. [...] AS does not move, since there is no change in current productivity."

a) (4 points) Redo the optimism experiment of §6.3 in the exam notation: $d\bar{a} > 0$, $da = 0$, i fixed. Curve, dy, dp, both gaps, the di that keeps $p = p^e$ and its cost.

Solution:

b) (4 points) Instead of $\bar{a}$, the news is that the long-run markup will *fall*, $d\bar{\mu} < 0$ (a competition reform that takes effect next period); μ does not change. Answer (a) for this shock and draw the AS–AD diagram with E_0, the new equilibrium E', the gap $y - y_n$, and the AD after the policy of price stability.

Solution:

c) (2 points) In words: what changes if the news is a *rise* in $\bar{\mu}$ (Prova 01, Q3)? Does the answer depend on ϕ?

Solution:

Question 3. (10 points) *Source: MPE Macroeconomics I, Lista 7 (2026), Question 2, item (a) only — reproduced verbatim (its points label and item (b) omitted); built also on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §7, Fig. 9, pp. 513–514, which has no exercises.*

Context from the source (added by this list, from Benigno 2015, pp. 507–508 and 513–514): The firms' "desired markup" of Lista 7 is μ in (YN), Benigno (15), p. 508. By (13), p. 507, μ combines the monopoly markup $\mu_\theta \equiv \theta/(\theta-1)$ (θ: elasticity of substitution among goods) with the tax rates on sales, labour costs, labour income and consumption. p. 513: a markup shock can come from more monopoly power ($\mu_\theta \uparrow$), higher tax rates, or a rise in the price of inputs with inelastic demand, such as oil. *Fig. 9* (temporary rise, $\mu_\theta \uparrow$): the natural level of output falls to y_n' while the efficient level y_e does not move; AS shifts up/left to AS$'$; AD does not move; the equilibrium moves from E to E', with lower output and higher prices ("stagflation"). Raising i shifts AD down to AD$''$ and reaches E'', with stable prices; lowering i shifts AD up to AD$'''$ and reaches E''', at the efficient level of output. There is a trade-off between the two objectives.

Lista 7, Question 2. Using separate AS–AD diagrams, analyze the following changes:

a) a temporary increase in firms' desired markups;

In each case, identify which curve shifts, indicate the direction of the shift, and describe the effects on output and the price level. Explain the economic intuition behind the movement of the curve and the resulting equilibrium.

What Benigno states (p. 513): "This section considers only temporary mark-up shocks, leaving other analyses to the reader."

Added parts (not in Lista 7), with the sign changed: the desired markup *falls* for one period only, $d\mu < 0$ (a temporary payroll-tax cut, or a fall in the oil price); $d\bar{\mu} = 0$. Items (a)–(c) below are this

list's.

a) (3 points) With i fixed: which curve moves and where; dy, dp, $d(y - y_n)$, $d(y - y_e)$ with signs.

Solution:

b) (3 points) Find the di that keeps $p = p^e$ and the di that keeps $y = y_e$. Report $y - y_e$ in the first case and $p - p^e$ in the second. Is there a trade-off?

Solution:

c) (4 points) The bank minimizes L. Find dy^*, dp^* and di^*, and the condition on ϕ under which it cuts.

Solution:

Question 4. (10 points) *Optimal policy. Sources: MPE Macroeconomics I, Lista 7 (2026), Question 4 — reproduced verbatim (its points label omitted); Carlin, W. and Soskice, D. (2024),* Macroeconomics: Institutions, Instability, and Inequality*, §3.6.1 checklist questions 4 and 5, p. 156 — reproduced verbatim; built also on Benigno, P. (2015), "New-Keynesian Economics: An AS–AD View", §11, eqs. (33)–(35), Fig. 14, pp. 521–523, which has no exercises.*

Context from the source (added by this list, from Lista 7, p. 1, and Benigno 2015, pp. 508 and 521): Lista 7 does not define y^*, ϕ_y, ϕ_p. y^* is the output level the bank targets (in Benigno's welfare-based loss (33), p. 521, below, the target is the efficient level, y_e); $\phi_y, \phi_p > 0$ are the weights on output deviations and on price surprises. The constraint is the AS of the cover, Benigno's (17), p. 508: p^e the preset price, y_n natural output, $\kappa > 0$ the slope. "The AS–AD dynamics seen in Benigno (2015)" are the cover's (AD) and (AS).

Lista 7, Question 4. The microeconomic foundations of the New Keynesian model make it possible to evaluate monetary policy in terms of household welfare.
Consider the following loss function in conjunction with the AS-AD dynamics seen in Benigno (2015)

$$\min_{y,p} \mathcal{L} = \phi_y(y - y^*)^2 + \phi_p(p - p^e)^2, \qquad \phi_y, \phi_p > 0,$$
$$\text{s.t:}$$
$$p - p^e = \kappa(y - y_n).$$

Interpret the shape of the loss function and analyse what it implies for the behavior of a central bank. Also, explain why this problem is subjected to the AS condition and what does that mean in terms of economic policy

Context from the source (added by this list, from Carlin and Soskice 2024, §3.2, pp. 129–132): The 3-equation model. Central bank loss function (p. 129):

$$L = (y_t - y_e)^2 + \beta(\pi_t - \pi^T)^2,$$

with y_t output, y_e equilibrium output (the level at which inflation is constant), π_t inflation, π^T the inflation target and β the relative weight on the loss from inflation ($\beta > 1$: an inflation-averse bank). Its indifference curves in the (y, π) plane are the "loss circles", centred on the "bliss point" (y_e, π^T), where $L = 0$; with $\beta = 1$ they are circles (p. 130, Fig. 3.5). Phillips curve, eq. (3.1), p. 131:

$$\pi_t = \pi_t^E + \alpha(y_t - y_e) = \pi_{t-1} + \alpha(y_t - y_e),$$

with expected inflation $\pi_t^E = \pi_{t-1}$ and slope $\alpha > 0$; each PC is labelled by its π_t^E, and the bank can only choose a point on the PC it faces. Monetary rule (MR), p. 132, which joins the tangencies between loss circles and Phillips curves:

$$y_t - y_e = -\alpha\beta(\pi_t - \pi^T).$$

Mapping to this list (ours): $\pi_t \leftrightarrow p$; π_t^E and $\pi^T \leftrightarrow p^e$; $\alpha \leftrightarrow \kappa$; C&S's y_e is both the anchor of the PC (the list's y_n) and the loss target (the list's y_e). Twice (LF) is $(y - y_e)^2 + \frac{\phi}{\kappa}(p - p^e)^2$, so $\beta \leftrightarrow \phi/\kappa$ and $\alpha\beta \leftrightarrow \phi$: the MR becomes $y - y_e = -\phi(p - p^e)$, Benigno's IT line below.

Carlin and Soskice, checklist questions 4 and 5.

4. Explain what is meant by the central bank's loss function. How are the central bank's preferences reflected in the loss function? Draw the loss 'circles' for the cases where

   (a) $\beta = 1$;
   (b) $\beta < 1$;
   (c) $\beta > 1$.

   In which of the three cases will the central bank reduce inflation back to target quickest after an inflation shock? Is there any downside to adopting this policy stance?

5. Assume that $\alpha = \beta = 1$, derive the MR curve graphically using the tangencies between the loss circles and the Phillips curves. With reference to the diagrams, explain the effect of the following (in each case, assume all other parameters are held constant):

   (a) An increase in the slope of the Phillips curve, α
   (b) An increase in central bank's inflation aversion, β.

Context from the source (added by this list, from Benigno 2015, §11, pp. 507 and 521–523): Welfare loss, a second-order approximation of household utility (p. 521):

$$L = \tfrac{1}{2}(y - y_e)^2 + \tfrac{1}{2}\tfrac{\theta}{\kappa}(p - p^e)^2, \tag{33}$$

with θ the elasticity of substitution among goods (p. 507; fn. 24, p. 522: θ around 8 gives a 15% markup). Substituting the AS, (17), and minimizing in y gives the optimal targeting rule (p. 522)

$$(y - y_e) + \theta(p - p^e) = 0, \tag{34}$$

called the IT equation: in the (y, p) plane its slope is $-1/\theta$ and it crosses (y_e, p^e); IT is flatter than AD whenever $\theta > \sigma$. General flexible inflation-targeting loss (p. 522), the cover's (LF):

$$L = \tfrac{1}{2}(y - y_e)^2 + \tfrac{1}{2}\tfrac{\phi}{\kappa}(p - p^e)^2, \tag{35}$$

with $\phi \geq 0$ the policymaker's preference; its rule is $(y - y_e) + \phi(p - p^e) = 0$, and a bank more concerned with price stability has a flatter IT. *Fig. 14*, p. 523 (temporary markup shock, $\mu_\theta \uparrow$): AS shifts to AS$'$ and, without intervention, the economy moves from E to E'; IT does not move, since y_e did not; the optimum is E'', where IT intersects AS$'$, reached by raising i (AD to AD$''$). p. 522: a bank less concerned with price stability, with IT steeper than AD, should lower i.

What Benigno states (p. 522): "This rule is robust to all types of shock and does not depend on their properties."

Added parts (not in the sources): items (a)–(c) below are this list's. They ask the questions above in Benigno's static model and with the loss L of the cover: κ plays the role of Carlin and Soskice's α, ϕ/κ of their β (so ϕ corresponds to $\alpha\beta$, as in the context box above), and the IT line of their MR curve.

a) (4 points) Minimize L subject to the AS. Derive the targeting rule, explain why the AS (and not the AD) is the constraint, and describe the loss contours and the IT line. Show that the Lista 7 loss $\phi_y(y - y^*)^2 + \phi_p(p - p^e)^2$ gives the same rule for suitable weights.

Solution:

b) (3 points) Let $W \equiv y_e - y_n > 0$. Find $y^* - y_e$ and $p^* - p^e$, sign their derivatives in ϕ and κ, and give the limits $\phi \to \infty$ and $\phi = 0$. Relate the answer to the two Carlin–Soskice questions.

Solution:

c) (3 points) For a shock with changes dy_n, dy_e, $d\bar{y}$ (g, $\bar{p}$, p^e fixed), find di^* and prove that it depends on ϕ if and only if $dy_n \neq dy_e$. Apply the result to news of a higher $\bar{\mu}$.

Solution:

Block 2 — Harder and conceptual: which bar moves which curve, signs, divine coincidence, zero bound

(Benigno §§4–11 • 38 points)

Question 5. (10 points) *Which bar moves which curve.* Consider a small rise, one at a time, in each of $a, \bar{a}, \mu, \bar{\mu}, g, \bar{g}, \bar{p}, p^e$.

a) (6 points) Build a table with, for each variable: the curve that moves and its shift (horizontal for AD and AS; vertical for p^e); whether y_e moves; whether there is a trade-off; and the di (sign and closed form) that keeps $y = y_n$ and $p = p^e$.

Solution:

b) (2 points) In words: state the rule that tells you which curve a variable moves.

Solution:

c) (2 points) In words: why is μ the only row with a trade-off?

Solution:

Question 6. (8 points) *Diagnose the answer (words only, no algebra).* News arrives that the long-run markup will rise ($d\bar{\mu} > 0$; current μ unchanged). A student writes four claims. For each, say whether it is right and, if not, correct it in words.

(1) "Firms will charge higher markups, so the AS shifts left."

(2) "Along the AD, prices go up and output goes down."

(3) "To keep prices stable the bank must raise i, accepting a deeper recession as the price of stable prices."

(4) "The optimal response depends on whether ϕ is larger than σ, i.e. on whether the IT line is flatter than the AD."

Solution:

Question 7. (9 points) *When does the divine coincidence fail?* True or false; justify each in words (1.5 points each).

Context from the source (added by this list, from Benigno 2015, pp. 507–509, 513–514 and 522): For (a): in the model only prices are sticky (a fraction of firms presets p^e); "An important assumption here is that of wage flexibility." (fn. 9, p. 513). For (c): the loss (33) of Question 4 is a second-order approximation of utility, and "The approximation holds around an efficient steady state in which $\mu = 0$." (fn. 21, p. 522). For (d): the consumption-tax rate τ_c (with τ_l, τ_w, τ_y the tax rates on labour income, labour costs and sales) enters the aggregate markup, $\mu \approx \mu_\theta + \tau_l + \tau_w + \tau_y + \tau_c$ (p. 514), hence (YN), Benigno (15); and it enters the AD, (21), p. 509:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma[i - (\bar{p} - p) - (\bar{\tau}_c - \tau_c) - \rho].$$

(YE), Benigno (19), p. 509, contains no tax rate.

a) If nominal wages were preset as well as prices, a productivity shock could create a trade-off between price stability and the efficient output gap.

b) A temporary rise in g forces the bank to choose between stable prices and a zero efficient gap.

c) Around a steady state with a large markup distortion, y_e need not be the welfare-relevant target, and the results for markup shocks are the ones most at risk.

d) A temporary rise in the consumption tax is a pure demand shock, so it creates no trade-off.

e) After bad news about $\bar{a}$, if the zero bound binds, the bank faces a trade-off between price stability and the efficient gap.

f) A dovish bank ($\phi < \sigma$) answers news of a higher $\bar{\mu}$ with a larger cut than a hawkish bank.

Question 8. (11 points) *The zero bound after $\bar{\mu}$ news.* Start from the steady state with $i = \rho > 0$. News arrives that the long-run markup will rise, $d\bar{\mu} > 0$; current μ is unchanged and the bank cannot set $i < 0$.

a) (3 points) Find the rate that keeps $p = p^e$ and the size $d\bar{\mu}^*$ above which the zero bound binds. Interpret with the natural rate.

Solution:

b) (3 points) Let $d\bar{\mu} > d\bar{\mu}^*$ and $i = 0$. Find dy, dp, $d(y - y_n)$, $d(y - y_e)$. Is there a trade-off?
Solution:

c) (3 points) Find the commitment $d\bar{p}$ that restores $y = y_n$ and $p = p^e$ with $i = 0$. Explain the channel and why credibility matters.
Solution:

d) (2 points) Alternatively, find the temporary dg that restores $p = p^e$ at $i = 0$, and say why fiscal policy works here.
Solution:

Block 3 — Created items: two shocks at once, and the timing of fiscal news

(created • 22 points)

Question 9. (12 points) *Created.* An oil-price rise lifts the current markup, $d\mu > 0$, and in the same week a law is passed that raises a payroll tax from next period on, $d\bar{\mu} > 0$. The bank minimizes L with a general $\phi \geq 0$.

a) (3 points) With i fixed: which curves move? Find dy, dp, $d(y - y_n)$, $d(y - y_e)$. When do prices rise?
Solution:

b) (4 points) Find dy^*, dp^*, di^* under the optimal rule. Split di^* into a part due to $d\bar{\mu}$ and a part due to $d\mu$; show which depends on ϕ; sign $\partial di^*/\partial\phi$; give the limits $\phi \to \infty$ and $\phi = 0$, and the condition for a cut.

Solution:

c) (2 points) Let the two shocks be equal, $d\bar{\mu} = d\mu$ (a permanent markup rise). Show that the optimal bank cuts for every ϕ, and explain why, although with i fixed prices do not move.

Solution:

d) (3 points) Draw the AS–AD diagram with $\sigma = 1$, $\eta = 0.5$ ($\Omega = 1.5$), $\kappa = 0.5$, $\phi = 3$, $d\mu = 1.5$, $d\bar{\mu} = 0.75$: AS, AS$'$, AD, AD$'$, the IT line, E_0, the no-policy point E', the optimum O with its AD, and both gaps at O. Give di^* and its two parts.

Solution:

Question 10. (10 points) *Created.* Public spending rises by $\Delta > 0$ in three ways: (i) temporarily, $dg = \Delta$, $d\bar{g} = 0$; (ii) permanently, $dg = d\bar{g} = \Delta$; (iii) as news only, $d\bar{g} = \Delta$, $dg = 0$.

a) (4 points) For each case: which curves move and by how much; dy, dp, $d(y - y_n)$ with i fixed; and $d(y - y_e)$.

Solution:

b) (3 points) Find the optimal di^* in each case and show that ϕ does not enter. Check that (i) and (iii) add up to (ii).

Solution:

c) (3 points) Draw the three diagrams for $\sigma = 1$, $\eta = \kappa = 0.5$, $\Delta = 1.5$. Then, in words: what is the output multiplier when the bank follows its optimal rule, and how does case (iii) relate to the $\bar{\mu}$ news of Prova 01?

Solution: